



Hidden Features of GWT

This article gives readers an overview of the hidden features of the Google Web Toolkit (GWT), which is used for developing AJAX-based Web applications in Java. It is aimed at those familiar with the basics of GWT.

Asynchronous JavaScript and XML (AJAX) is not a language or new technology, but a combination of known technologies such as HTML, XML and JavaScript. It has been combined to serve users in a better and faster way. Being asynchronous is the key; all user actions are handled in parallel. In traditional Web applications, all user actions were handled sequentially. AJAX requires HTML for creating and loading pages, JavaScript for altering page content, XML/JSON for data communication, and server-side business logic. AJAX-based Web applications reduce unnecessary server calls and the reloading of the entire page for small user requests or actions.

The Google Web Toolkit (GWT)

GWT is an open source Java-based framework for developing AJAX-based Web applications. A Java-based framework like Swing, it enables developers to develop rich Web applications without any knowledge of HTML/JavaScript. GWT provides solutions for known drawbacks of AJAX, which are listed below:

- GWT provides a compiler, which converts all code written in Java into JavaScript. Because of this, developing applications with cross-browser compatibility becomes easier.
- It has a way of maintaining a history of accessed application URLs, letting you use the browser's *Forward/Back* button.
- Debugging of client-side code is possible. Since we are developing in Java, debugging the AJAX application is possible during development, using a native IDE.

The internals of GWT

The simple client-side architecture of a GWT application is shown in Figure 1.

The Java source code in the client source package will be compiled to JavaScript by the GWT compiler, so the client side of the application is restricted to using classes from the GWT widgets library, and other modules and classes supported by the GWT JRE emulation library. This library supports part of the classes from *java.lang*, *java.io* and *java.util*. GWT